



STM32C5 microcontroller series

Upgrade your entry-level design
without raising your budget





BREAKING
THE
MOLD
WITHOUT
BREAKING
THE
BANK



STM32C5 



The STM32 portfolio

Five product categories



Wireless
MCU

Short- and long-range connectivity



Ultra-low-power
MCU

32-bit general-purpose microcontrollers: from 75 to 5,072 CoreMark score



Mainstream
MCU



High-performance
MCU



Embedded
MPU

32- and 64-bit microprocessors



Enabling edge AI solutions



Scalable security



[MPU portfolio](#)
[MCU portfolio](#)



What the STM32C5 series offers



**Enhanced performance for
developers and end device
users**



**Without impacting
end-product cost**

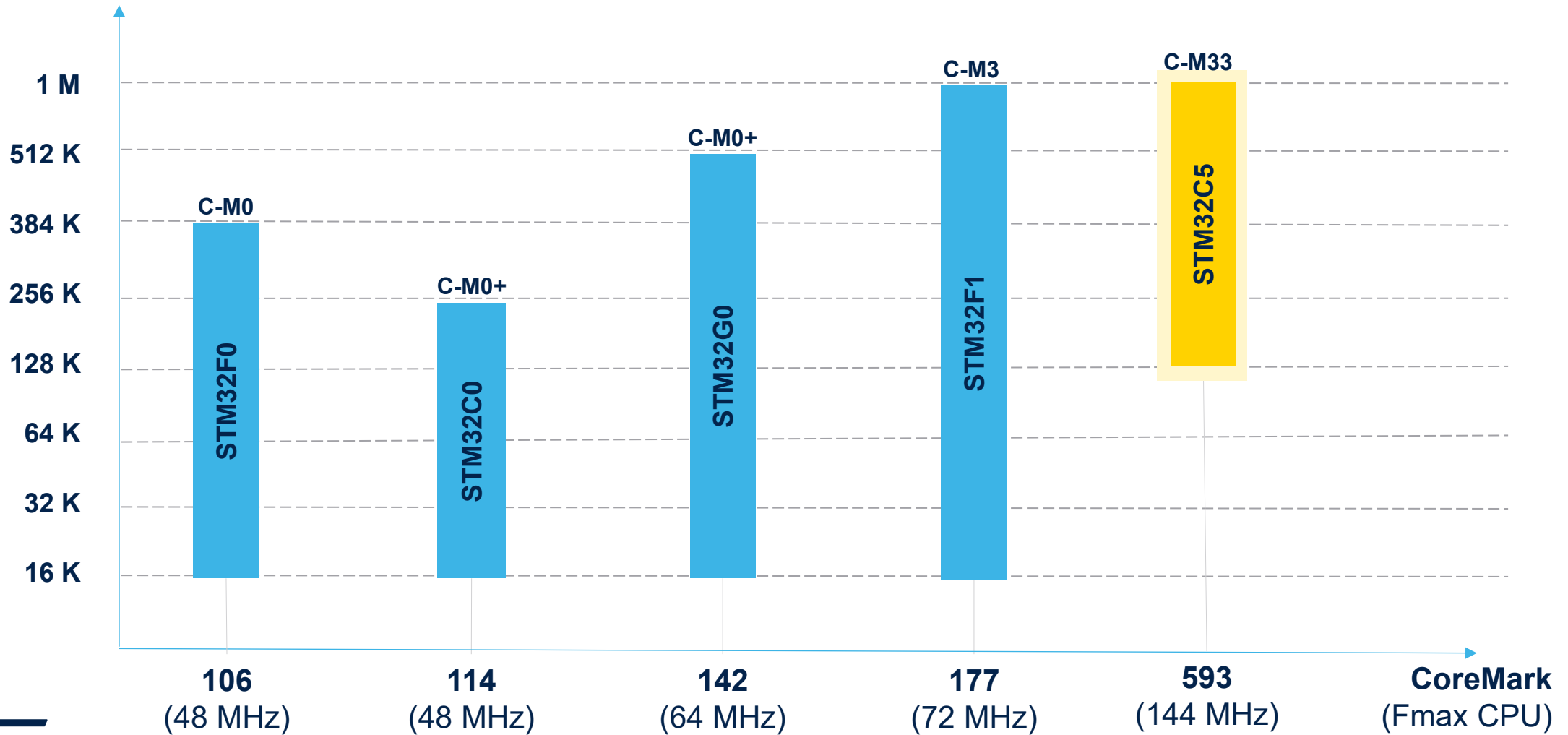


**Building on the foundation
of the STM32 offering**



New STM32C5 series

Flash memory size (bytes)



Expanding the potential of entry-level applications



Smart homes & appliances

Home appliances, thermostats, smoke detectors, door locks, sensors



Factory automation

Sensors, controllers, robotics, PLCs



Entertainment

Gaming accessories, keyboards

A modern platform

Built on Arm® Cortex® -M33 core

Powered by ST's 40nm process

Enabling a smooth design journey

- **No external accelerators required**
 - Integrated floating-point unit simplifies software development & hardware division
 - Enhanced native DSP and math instructions
- Easier peripheral separation and software management thanks to **more flexible** interrupt vectors
- **Easier debugging** with improved trace features
- Boosted edge AI algorithms
- Cortex® -M33 is **supported by all major compilers**

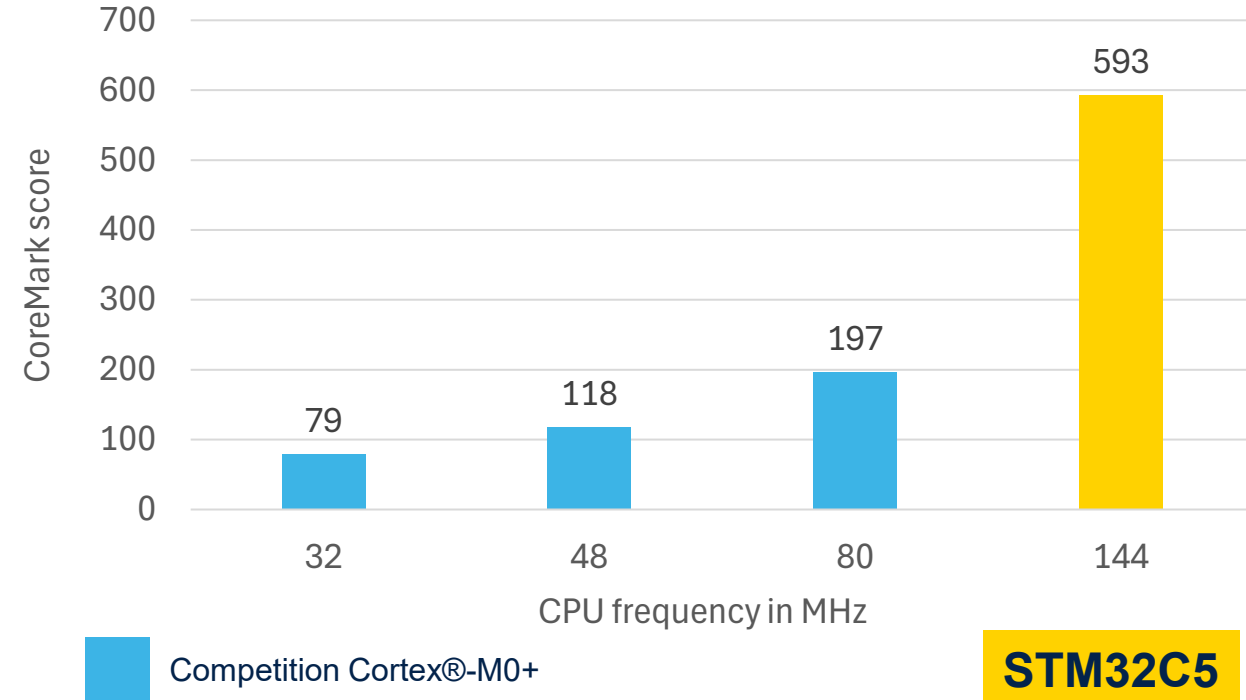


512 Kbytes of flash, 128 Kbytes of SRAM,
64 pins, starts at \$0.87 for 10k units

A modern core providing a performance leap for the entry range

	Cortex®-M0+	Cortex®-M33
Instruction set	Arm V6	Arm V8
FPU	-	Yes
Hardware divide	-	Yes
DSP instructions	-	Yes
CoreMark/MHz	2.46	4.12
FIR filter (float32) perf	Reference	40x more performance
FIR filter (int32) perf	Reference	20x more performance
Flexible interrupt vectors	Up to 32 inputs	Up to 480 inputs

STM32C5 provides **three times more performance** than the competition Cortex®-M0+ in the market



Reliable performance in industrial environments



A robust platform and precise clock system

- Arm Cortex®-M33 core up to **144 MHz** with FPU, MPU, DSP, and flexible NVIC
- Up to **125°C** ambient temperature without performance penalty
- **1%** factory-calibrated internal oscillator between wide ambient temperature range from **-20 to 105°C**
- Multiple external clock sources including high-speed (HSE), low-speed (LSE), and internal oscillators for flexibility and redundancy
- Versatile analog features available (op amp, DAC, 12-bit ADC, comparator)

Industrial interfaces and flexible memory

- More RAM available compared to typical Cortex-M0 +-based applications
- **OctoSPI** for external memory expansion
- Industrial-grade communication interfaces: **Ethernet, FDCAN, I3C**

Security & safety

- RDP and password-protected regression
- Hardware security with side channel attack resistant crypto
- PSA Certified Level 3 & SESIP3 target certifications
- Ready for functional safety requirements



Certified functional safety packages for STM32C5



www.st.com/x-cube-stl

Robust performance and reliability in harsh conditions

- ECC on RAM and flash
- Brown-Out Reset
- Dual Watchdog

SIL functional safety

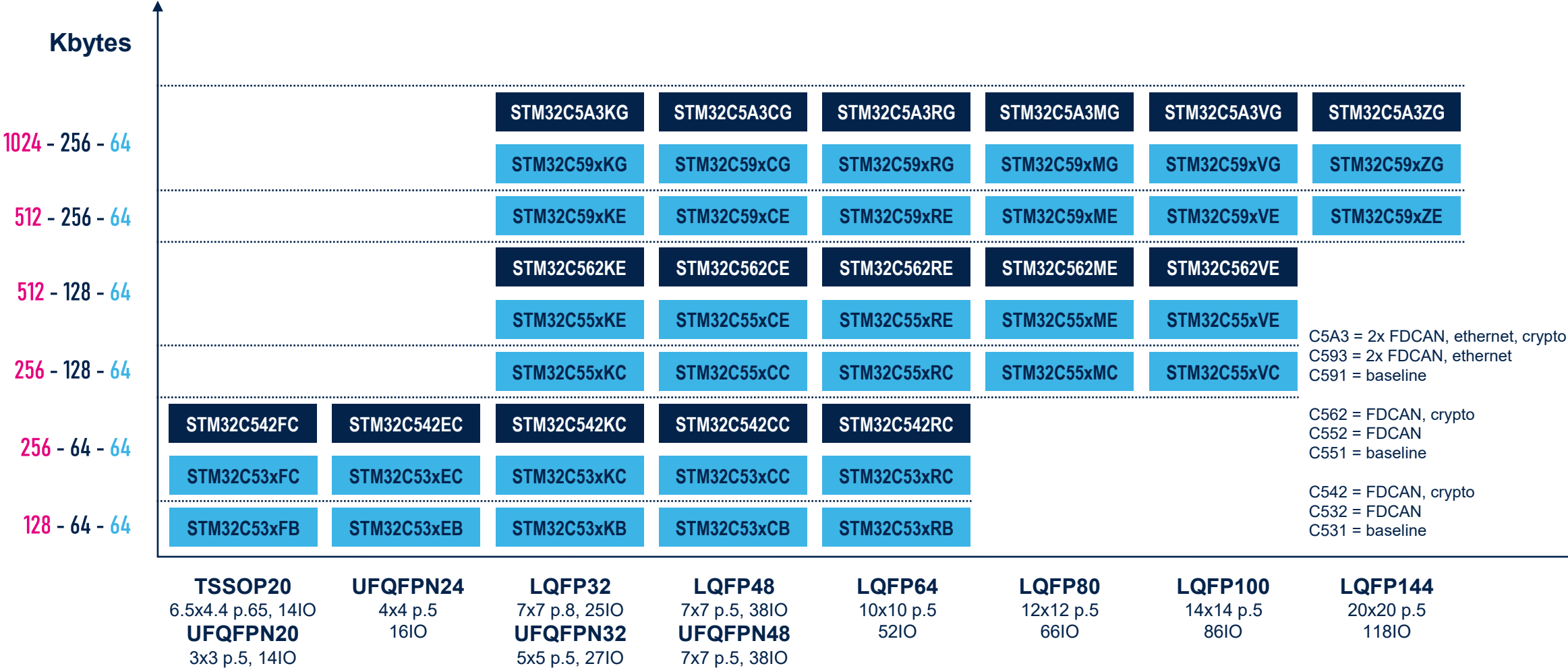
- Ready for IEC 61508 systems up to SIL3
- FMEDA & FMEA, safety manual
- Certified STM32 Self-Test library

Class B functional safety

- IEC 60335-1/60730-1 Class-B
- Household electrical appliances
- Certified STM32 Self-Test library

One new STM32 series. Multiple lines.

Flash - RAM - Flash area for software EEPROM emulation



Commercial part number availability in $T_{\text{ambient}} = -40$ to $85/125^{\circ}\text{C}$

Crypto line

Non-crypto

STM32C59x/A3 block diagram

Arm® Cortex®-M33
144 MHz
593 CoreMark

FPU, MPU, DSP, ETM
 SW debug

8 KB - ICACHE

Memory

1 MB flash
 Dual Bank w/ ECC

256 KB SRAM w/ ECC

64 KB flash area for
 software emulated
 EEPROM

4.5 KB
 OTP

128B bkp
 RAM

System

VDD 2.7-3.6V

32kHz LSE, LSI, HSE
 144MHz HSI with internal
 1% accuracy Ta=-20-105°C

RTC

Connectivity

4x USART, 3x UART
 1x LPUART

3x SPI

2x I2C, 1x I3C

1x USB FS

2x FDCAN ⁽²⁾

1x OCTOSPI

ETH 10-100Mb ⁽²⁾

Security

SAES, AES 128/256 ⁽¹⁾

TRNG

HASH, OTP, HUK
 SPKA, CCB

96-bits unique ID

Accelerators

2x DMA (8+8)

CORDIC, CRC

Analog

3x 12-bit ADC
 2.25MSPS

1x DAC12
 w/ Buffer

1x COMP

Timers

4x 32-bit GPTIM

2x 16-bit 7ch MCTIM

4x 16-bit GPTIM

2x 16-bit basic TIM

1x 16-bit LPTIM

IWDG, WWDG
 SysTick

Operation modes

Run, Sleep, STOP,
 Standby

Bootloader

UART, SPI, FDCAN, USB

Operating temperature

T_{ambient} -40 to 125°C

T_{junction} 140°C

Packages

QFP (0.5p) 48, 64, 80, 100,
 144
 QFN (0.5p) 32, 48

Cortex-M33 for all entry products

Extensive memory

More connectivity

(1) Available on STM32C5A3 lines

(2) Available on STM32C593 and STM32C5A3 lines

Find the right fit for your design thanks to a scalable portfolio

	STM32C531	STM32C532	STM32C542	STM32C551	STM32C552	STM32C562	STM32C591	STM32C593	STM32C5A3
Flash size with ECC	128 to 256 KB	128 to 256 KB	256 KB	256 to 512 KB	256 to 512 KB	512 KB	512 KB to 1 MB	512 KB to 1 MB	1 MB
RAM size with ECC	64 KB	64 KB	64 KB	128 KB	128 KB	128 KB	256 KB	256 KB	256 KB
12-bit ADC	1	1	1	2	2	2	3	3	3
DAC/COMP	2/2	2/2	2/2	1/1	1/1	1/1	1/1	1/1	1/1
OPAMP	1	1	1						
Memory IF							1x OctoSPI	1x OctoSPI	1x OctoSPI
FDCAN		2	2		1	1		2	2
Ethernet 10/100Mbit								Yes	Yes
AES			Yes			Yes			Yes
HASH	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SAES/SPKA, HUK, SCA, CCB							SPKA verify	SPKA verify	Full config



STM32C53x/42 block diagram

Arm® Cortex®-M33
144 MHz
593 CoreMark

FPU, MPU, DSP, ETM
SW debug

8 KB - ICACHE

Memory

256 KB flash
Dual Bank w/ ECC

64 KB SRAM w/ ECC

64 KB flash area for
software emulated
EEPROM

4.5 KB OTP	128B bkp RAM
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System

VDD 2.7-3.6V

32kHz LSE, LSI, HSE
144MHz HSI with internal
1% accuracy Ta=-20-105°C

RTC

Connectivity

2x USART, 2x UART
1x LPUART

2x SPI

2x I2C, 1x I3C

1x USB FS

2x FDCAN ⁽²⁾

Security

AES 128/256 ⁽¹⁾

TRNG

HASH, OTP

96-bits unique ID

Accelerators

DMA (4+4)

CORDIC, CRC

Analog

1x 12-bit ADC
2.25MSPS

2x DAC12
w/ Buffer

2x COMP

1x OPAMP

Timers

1x 32-bit GPTIM

2x 16-bit 7ch MCTIM

2x 16-bit GPTIM

2x 16-bit basic TIM

1x 16-bit LPTIM

IWDG, WWDG
SysTick

Operation modes

Run, Sleep, STOP,
Standby

Bootloader

UART, SPI, FDCAN, USB

Operating temperature

T_{ambient} -40 to 125°C

T_{junction} 140°C

Packages

QFP (0.5p) 32, 48, 64
QFN (0.5p) 20, 24, 32, 48
TSSOP20

Cortex-M33 for all entry products

Extensive memory

(1) Available on STM32C542 lines

(2) Available on STM32C532 and STM32C542 lines

STM32C55x/62 block diagram

Arm® Cortex®-M33
144 MHz
593 CoreMark

FPU, MPU, DSP, ETM
 SW debug

8 KB - ICACHE

Memory

512 KB flash
 Dual Bank w/ ECC

128 KB SRAM w/ ECC

64 KB flash area for
 software emulated
 EEPROM

4.5 KB
 OTP

128B bkp
 RAM

System

VDD 2.7-3.6V

32kHz LSE, LSI, HSE
 144MHz HSI with internal
 1% accuracy Ta=-20-105°C

RTC

Connectivity

3x USART, 2x UART
 1x LPUART

3x SPI

2x I2C, 1x I3C

1x USB FS

1x FDCAN ⁽²⁾

Security

AES 128/256 ⁽¹⁾

TRNG

HASH, OTP

96-bits unique ID

Accelerators

2x DMA (8+4)

CORDIC, CRC

Analog

2x 12-bit ADC
2.25MSPS

1x DAC12
 w/ Buffer

1x COMP

Timers

2x 32-bit GPTIM

2x 16-bit 7ch MCTIM

4x 16-bit GPTIM

2x 16-bit basic TIM

1x 16-bit LPTIM

IWDG, WWDG
 SysTick

Operation modes

Run, Sleep, STOP,
 Standby

Bootloader

UART, SPI, FDCAN, USB

Operating temperature

T_{ambient} -40 to 125°C

T_{junction} 140°C

Packages

QFP (0.5p) 32, 48, 64, 80,
 100
 QFN (0.5p) 32, 48

Cortex-M33 for all entry products

Extensive memory

(1) Available on STM32C562 lines

(2) Available on STM32C552 and STM32C562 lines

Security on STM32C5



Choose your STM32C5 based on your security strategy



Advanced

Aim for a higher security level
with side channel attack resistant crypto

STM32C59x/5A3

Focus on the essentials

Enable custom software security
by using hardware features

STM32C55x/562
STM32C53x/542

Increasing security with digital identities

Memory protections against illegal access control	Cryptography for hardware robustness	Security services	
OTP, HDP, WRP, RDP, MPU Passive tamper	(*) SAES, SPKA, SHA512 SHA, SHA1, SHA224/256 TRNG Hardware unique key (HUK)	Secure boot & secure updates	
Platform protection during product lifecycle	Code isolation for runtime protection	Secure KeyStore with CCB*	
RDP: 2 protection level states Password based regression	Temporal isolation – HDP	NIST - CAVP certified X-CUBE-CRYPTOLIB (**) Flexible Crypto Framework	

State-of-the-art security assurance level



psacertified™
level three

target certifications

Robust security to safeguard sensitive and mission-critical applications

More locked doors to ensure device security on STM32C59x/5A3

Enhanced device authentication and anticloning capabilities at a reduced cost

#1

The coupling and chaining bridge (**CCB**) HAL

- Isolated Secure crypto engines with private key bus to chain crypto operations without involving CPU.
- Keeps keys hidden even from the CPU and securely stored.

#2

Secure key storage

- Key wrapping with hardware unique key and CCB.
- Key isolation during life cycle with HiDeProtect.



Side-channel resistant hardware protection

Strong security for regulatory compliance, including CRA

SESIP3 and PSA Certified Level 3 Certification targets



Tailored for entry-level graphics



STM32 hardware for entry-level graphics

Graphic performance	ENTRY LEVEL							
Display	SPI/Quad/OctoSPI and parallel display IF, G-RAM, touch, and no-touch							
STM32 MCU	STM32U0	STM32U3	STM32C0	STM32G0	STM32H503	STM32C5	STM32WBA	STM32WB
Development kits- TouchGFX supported	Nucleo-U083RC + X-NUCLEO-GFX01M2	Nucleo-U385RG-Q + RVA15MD*	Nucleo-C092RC + RVA15MD*/ X-NUCLEO-GFX01M2	Nucleo-G071RB + X-NUCLEO-GFX01M2	Nucleo-H503RB + X-NUCLEO-GFX01M2	NUCLEO-C5A3ZG + RVA15MD	Nucleo-WBA65RI + RVA15MD*	Nucleo-WB55RG + X-NUCLEO-GFX01M2
Main features <i>Max memory listed</i>	CM0+, 56 MHz 40 KB RAM 256 KB flash	CM33, 96 MHz 256 KB RAM 1000 KB flash	CM0+, 48 MHz 36 KB RAM 256 KB flash	CM0+, 64 MHz 144 KB RAM 512 KB flash	CM33, 250 MHz 32 KB RAM 128 KB flash	CM33, 144 MHz 64-256 KB RAM 128-1024 KB flash	CM33, 100 MHz 512 KB RAM 1000 KB flash	CM4, 64 MHz 256 KB RAM 1000 KB flash
Framebuffer details	Partial in internal RAM, starting from 6 KB					Full or partial in internal RAM		
Asset storage (flash memory)	Internal or External SPI	Internal or Externa(Q)SPI	Internal or External SPI	Internal or External SPI	Internal or Externa(Q)SPI	Internal or External (O)SPI	Internal or External SPI	Internal or Externa(Q)SPI

*Riverdi add-on kit: RVA15MD, order from Riverdi sales channels: [1.54" STM32 Nucleo-64 display kit - Riverdi](#)



**Achieve STM32C5 design success with
STM32Cube ecosystem**



STM32C5 tailored for entry-level GUIs

STM32 hardware

STM32 software



All STM32C5 Nucleo kits



Extension display from Riverdi



One-chip module from TSD



TouchGFX +



A complete STM32 software offer,
free of charge

Helping developers release their creativity

Develop faster

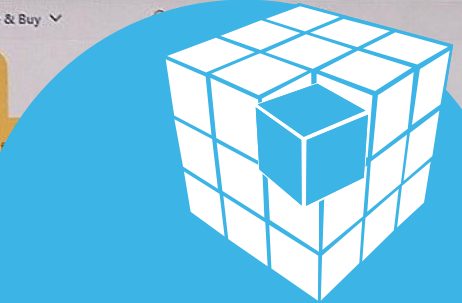
With STM32 code generation tools, IDE, and production-ready examples

Stand out more

With optimized drivers leveraging all STM32 smart features

Save costs

With free and permissive production licenses

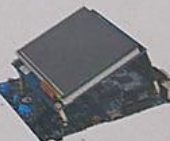


STM32 Cube

STM32 MCU b

Evaluate, load, and debug v

ware tools Evaluation and expansion boards



boards
valuation and prototyping

STM32 Discovery kits
Feature-rich evolution and prototyping

STM32 Eval boards
Full feature evaluation

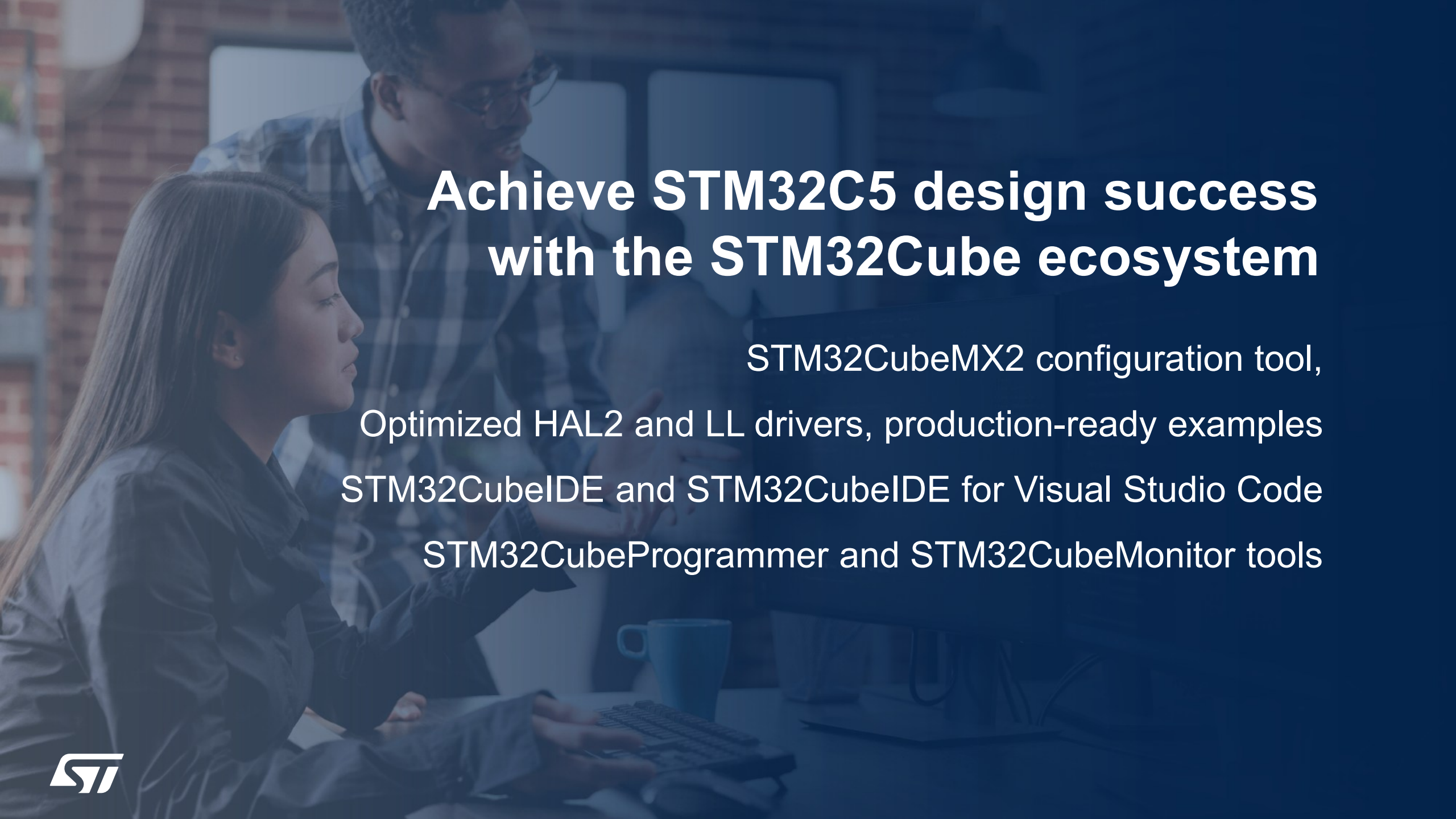


Nucleo expansion boards
Using ST product technologies

Accessories for boards
Daughter boards and adapters

Debuggers & programmers
For ST and partner tools



A man and a woman are working together at a computer in an office setting. The man is standing and pointing at the screen, while the woman is sitting and looking at the screen. The background is a blurred office environment with a brick wall and a window.

Achieve STM32C5 design success with the STM32Cube ecosystem

STM32CubeMX2 configuration tool,

Optimized HAL2 and LL drivers, production-ready examples

STM32CubeIDE and STM32CubeIDE for Visual Studio Code

STM32CubeProgrammer and STM32CubeMonitor tools



Takeaway

- Starting at \$0.64USD at 10ku
- Reduced BOM cost
- Offers exceptional value/price compared to typical M0+-based MCUs



- Rich analog capability: up to 3 ADC, 2 comparators, high analog content, 3 ADCs 1 OPAMP
- Broad connectivity: Ethernet 10/100, 2 FDCAN, USB, OCTOSPI
- HASH, HUK, CCB, Side Channel attack protected PKA and AES, SESIP3 & PSA certified level 3 target certifications
- Flexible DMA for efficient data handling

- Offers low dynamic power consumption with $<80 \mu\text{A}/\text{MHz}$
- Integrates a 1% calibrated RC oscillator across -20 to 105°C temperature
- Maintains full performance across the entire temperature range
- Available in wide packages from 20 to 144 pins, starting at 3x3mm

